will still come to a consensus on the next block. In BigchainDB, Byzantine fault tolerance is achieved using Tendermint protocols.

- **Immutable storage:** In BigchainDB, if a node gets corrupted or some data is changed, then it is detectable. Because, every node has a local copy of the MongoDB database and transactions are cryptographically signed.
- **Owner-controlled assets:** BigchainDB allows users to create their own private network with custom assets and required permissions.
- **Query engine:** Each node has its own local MongoDB database resulting in powerful query functionality.
- **Low latency:** All the communications between nodes are completed using Tendermint protocols and thus can process a large number of transactions per second.
- **Open source:** Being open source with full community backing, everyone can use it and build their own application on top of the BigchainDB stack.

Features	Blockchain	Distributed database	BigchainDB
Decentralisation	✔		✔
Byzantine fault tolerance	✔		✔
Immutability	✔		✔
Owner-controlled assets	✔		✔
High transaction rate		✔	✔
Low latency		✔	✔
Open source	✔	✔	✔

Table 1: BigchainDB combines the key benefits of distributed DBs and blockchain

Key differences between MySQL, a traditional database, and BigchainDB

Traditional databases like MySQL provide centralised systems that cannot satisfy the requirements of blockchain networks. This is where BigchainDB is used. The key differences between traditional databases like MySQL and the BigchainDB blockchain database are shown in Table 2.

How BigchainDB works?

How BigchainDB works can be understood by following the life cycle of a transaction. A BigchainDB transaction is a JSON string that conforms to a BigchainDB transaction specification. The system architecture of the BigchainDB network is shown in Figure 1, and it consists of four main components – the BigchainDB node, the BigchainDB server, Tendermint and the MongoDB database. The steps of a BigchainDB transaction are described below.

***Step 1:** Sends the transaction to a BigchainDB network:* When a user initiates the transaction, the BigchainDB HTTP API is used to send the transaction to a BigchainDB network.

***Step 2:** Arrival of the transaction at BigchainDB server:* Once the transaction is successfully received and validated at the BigchainDB server, it is converted into Base64 and a new JSON string. The BigchainDB server then sends this JSON string to a local Tendermint node.

***Step 3:** Arrival of the transaction at a Tendermint instance:* When the transaction arrives at the Tendermint node, it is processed using the CheckTx API and included in the

Features	MySQL	BigchainDB
Purpose	Used for data warehousing, logging and e-commerce applications	Used for blockchain technology
Database model	RDBMS	JSON
Licence	Open source	Open source
Data schema	Yes	Schema-free
Performance overhead	Bottlenecks can occur with high traffic	Bottlenecks cannot occur
Accessibility	Limited access by one or more person to the same set of data	Most data is local and in close proximity to where it is required
APIs and other access methods	ADO.NET, JDBC, ODBC and proprietary native APIs	CLI-client, Restful HTTP API

Table 2: Key differences between MySQL and BigchainDB